

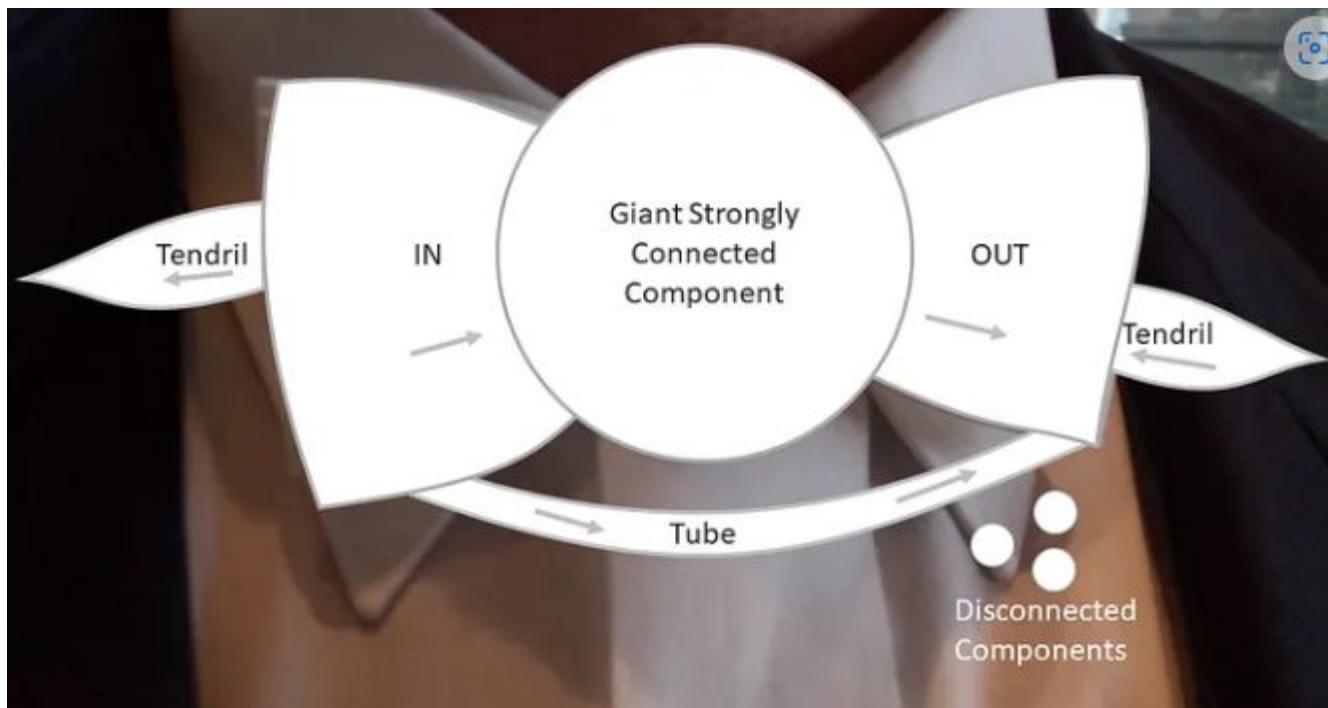
Web Information Retrieval

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The World Wide Web is like a bow tie

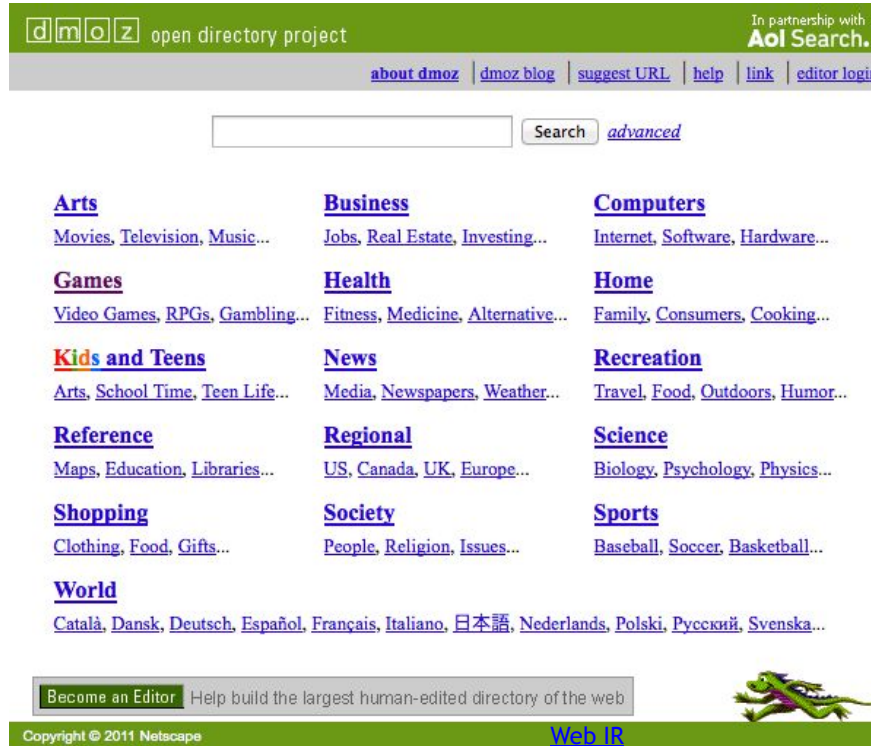


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Hierarchical Category Scheme Example

DMOZ



The screenshot shows the DMOZ homepage with a green header bar. The header contains the DMOZ logo, the text "open directory project", and a partnership with AOL Search. Below the header is a navigation bar with links: "about dmoz", "dmoz blog", "suggest URL", "help", "link", and "editor login". A search bar is located below the navigation bar, with a "Search" button and a link to "advanced". The main content area displays a grid of category links, each with a sub-link list. The categories are: Arts (Movies, Television, Music...), Business (Jobs, Real Estate, Investing...), Computers (Internet, Software, Hardware...), Games (Video Games, RPGs, Gambling...), Health (Fitness, Medicine, Alternative...), Home (Family, Consumers, Cooking...), Kids and Teens (Arts, School Time, Teen Life...), News (Media, Newspapers, Weather...), Recreation (Travel, Food, Outdoors, Humor...), Reference (Maps, Education, Libraries...), Regional (US, Canada, UK, Europe...), Science (Biology, Psychology, Physics...), Shopping (Clothing, Food, Gifts...), Society (People, Religion, Issues...), Sports (Baseball, Soccer, Basketball...), and World (Català, Dansk, Deutsch, Español, Français, Italiano, 日本語, Nederlands, Polski, Русский, Svenska...). At the bottom, there is a "Become an Editor" button and a link to "Help build the largest human-edited directory of the web". A small green lizard logo is also present in the bottom right corner.

Arts
[Movies](#), [Television](#), [Music](#)...

Business
[Jobs](#), [Real Estate](#), [Investing](#)...

Computers
[Internet](#), [Software](#), [Hardware](#)...

Games
[Video Games](#), [RPGs](#), [Gambling](#)...

Health
[Fitness](#), [Medicine](#), [Alternative](#)...

Home
[Family](#), [Consumers](#), [Cooking](#)...

Kids and Teens
[Arts](#), [School Time](#), [Teen Life](#)...

News
[Media](#), [Newspapers](#), [Weather](#)...

Recreation
[Travel](#), [Food](#), [Outdoors](#), [Humor](#)...

Reference
[Maps](#), [Education](#), [Libraries](#)...

Regional
[US](#), [Canada](#), [UK](#), [Europe](#)...

Science
[Biology](#), [Psychology](#), [Physics](#)...

Shopping
[Clothing](#), [Food](#), [Gifts](#)...

Society
[People](#), [Religion](#), [Issues](#)...

Sports
[Baseball](#), [Soccer](#), [Basketball](#)...

World
[Català](#), [Dansk](#), [Deutsch](#), [Español](#), [Français](#), [Italiano](#), [日本語](#), [Nederlands](#), [Polski](#), [Русский](#), [Svenska](#)...

[Become an Editor](#) Help build the largest human-edited directory of the web

Copyright © 2011 Netscape [Web IR](#)

Browsing

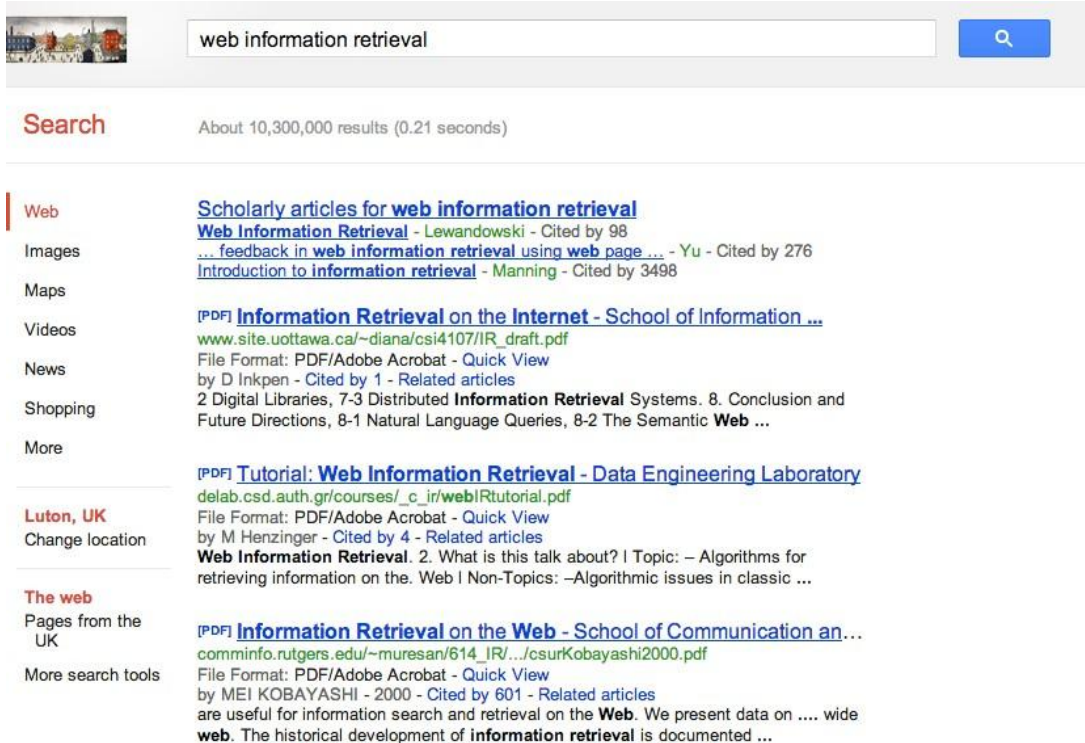
- Document search can be simplified by navigating to interesting/relevant categories
- Catalogues like (the now defunct) *Yahoo!* and DMOZ offered a
- **hierarchical category scheme**
- Each Web document in this catalogue is assigned to one or more categories, which gives useful hints about the document topic

Topic gets more specific the deeper we get

Searching

- The vast amount of Web documents cannot be made available without search technology
- “The Web is a jungle” (Sergey Brin)
- Often there is an interplay between searching and browsing We can apply standard IR methods (e.g. VSM, inverted lists, $tf \times idf$) plus **link analysis**

Anatomy of a Web Search Engine:



The screenshot shows a Google search interface. At the top, there's a search bar with the text "web information retrieval" and a magnifying glass icon. Below the search bar, it says "Search" and "About 10,300,000 results (0.21 seconds)". On the left side, there's a vertical menu with categories: Web, Images, Maps, Videos, News, Shopping, and More. The "Web" category is selected. The main content area displays search results. The first result is titled "Scholarly articles for web information retrieval" and lists several articles with their authors and citation counts. The second result is titled "[PDF] Tutorial: Web Information Retrieval - Data Engineering Laboratory" and provides a link to a PDF file. The third result is titled "[PDF] Information Retrieval on the Web - School of Communication an..." and also provides a link to a PDF file. The search results are formatted with blue links and red text for the title and author information.

Search About 10,300,000 results (0.21 seconds)

Web

[Scholarly articles for web information retrieval](#)
[Web Information Retrieval](#) - Lewandowski - Cited by 98
... feedback in [web information retrieval](#) using [web page](#) ... - Yu - Cited by 276
[Introduction to information retrieval](#) - Manning - Cited by 3498

[\[PDF\] Information Retrieval on the Internet - School of Information ...](#)
[www.site.uottawa.ca/~diana/csi4107/IR_draft.pdf](#)
File Format: PDF/Adobe Acrobat - Quick View
by D Inkpen - Cited by 1 - Related articles
2 Digital Libraries, 7-3 Distributed **Information Retrieval** Systems. 8. Conclusion and
Future Directions, 8-1 Natural Language Queries, 8-2 The Semantic **Web** ...

[\[PDF\] Tutorial: Web Information Retrieval - Data Engineering Laboratory](#)
[delab.csd.auth.gr/courses/_c_ir/webIRtutorial.pdf](#)
File Format: PDF/Adobe Acrobat - Quick View
by M Henzinger - Cited by 4 - Related articles
Web Information Retrieval. 2. What is this talk about? | Topic: – Algorithms for
retrieving information on the. Web | Non-Topics: –Algorithmic issues in classic ...

Luton, UK
Change location

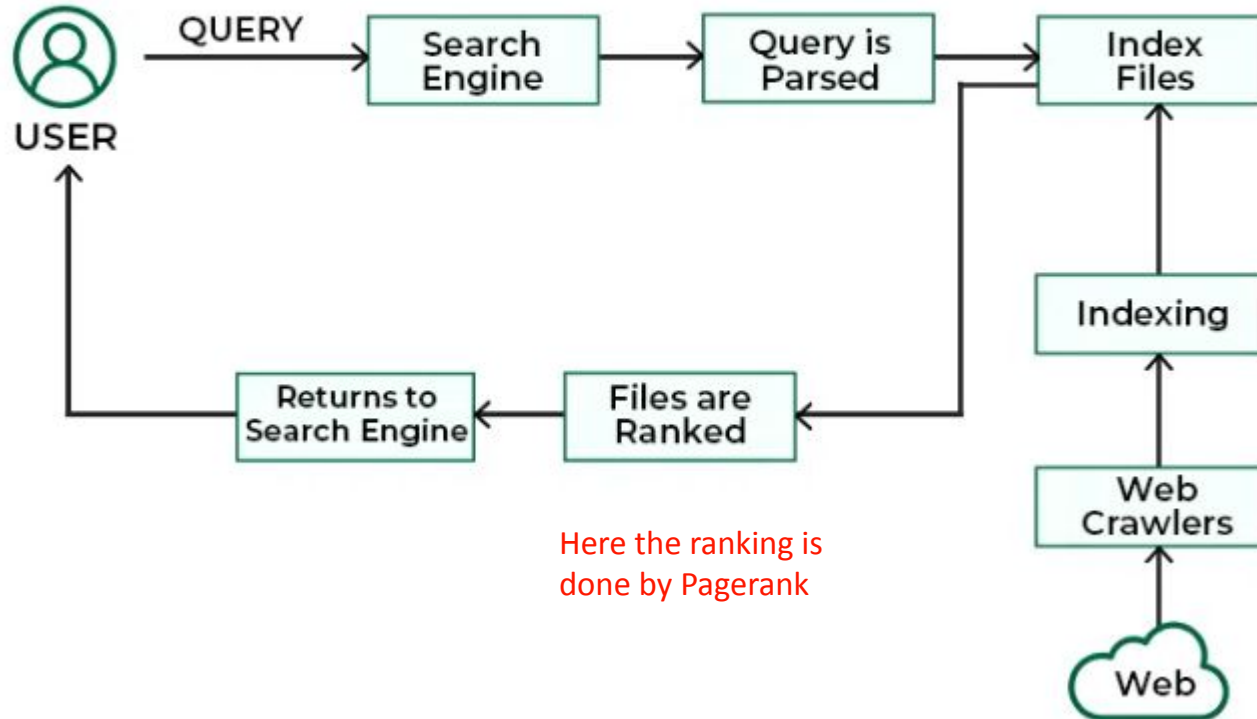
The web

[\[PDF\] Information Retrieval on the Web - School of Communication an...](#)
[comminfo.rutgers.edu/~muresan/614_IR/.../csurKobayashi2000.pdf](#)
File Format: PDF/Adobe Acrobat - Quick View
by MEI KOBAYASHI - 2000 - Cited by 601 - Related articles
are useful for information search and retrieval on the **Web**. We present data on wide
web. The historical development of **information retrieval** is documented ...

Example: Google



Architecture of Search Engine



Here the ranking is
done by Pagerank

Collects HITs
|
Stores them in
Barrel

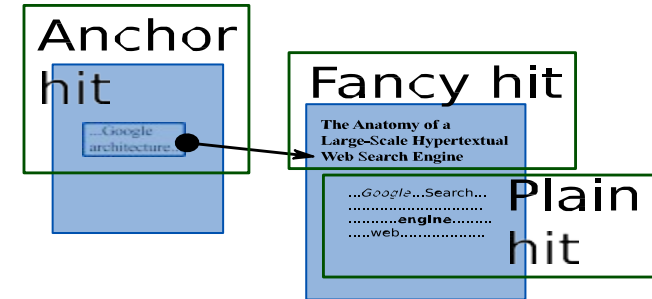
Google Hit Types

Google not only looks for the occurrence of a term (or word), but also separates between 3 different types of occurrences (called **Hits**):

Fancy Hit Includes hits occurring in a URL, title, or meta tag

Anchor Hit Special kind of fancy hit, occurrence in an anchor text linking to the page (anchor propagation)

Plain Hit All the rest



Link Analysis with PageRank and HITS

Web Information Retrieval

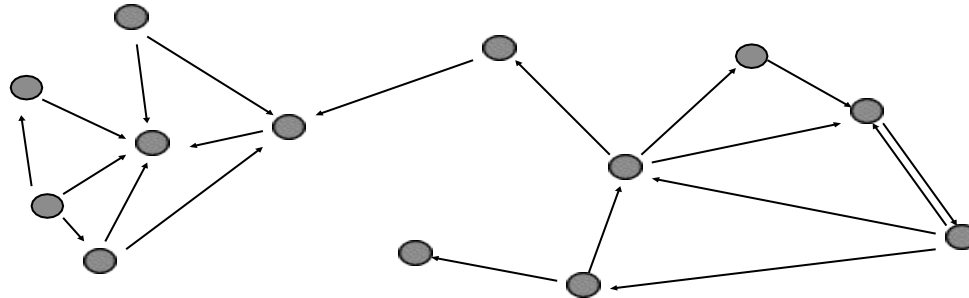
- Search engines apply known IR techniques (like inverted lists, $tf \times idf$)
- The approaches so far were just content-based (computing the retrieval status value (RSV) w.r.t. a query, rank documents w.r.t. decreasing RSV)
- On the Web we have another source of information: the **link structure**
 - As we've seen, Google stores more than an inverted index, but also information about the link structure

Performing a **link analysis**, we can extract from the link structure some further information going beyond mere document content

The Web as a Hypertext

Definition (Hypertext)

A **Hypertext** is a set of vertices (nodes) and directed edges (links) forming a graph. The Web can be regarded as a cyclic graph.



Hypertext and Web Information Retrieval

Hypertext IR uses the node content and the link structure for Information Retrieval

In that sense Web IR is a special case of Hypertext IR

In fact, Hypertext IR has a longer history than Web IR

Possibilities include:

Identify the “importance” of documents and of hubs and authorities (**PageRank**, **HITS**)

PageRank

[[Page et al., 1998](#)]

- Famous Google approach
- Determines the **authority** of a document (or page)
- **Assumption**: a document is more authoritative, the more other documents link to it
- *Iterative* calculation of the PageRank by propagating the PageRank to successor nodes

Random Surfer Model

The assumption behind Page Rank is that of a **random surfer**.

Random Surfer Model



- Random walks on the Web graph

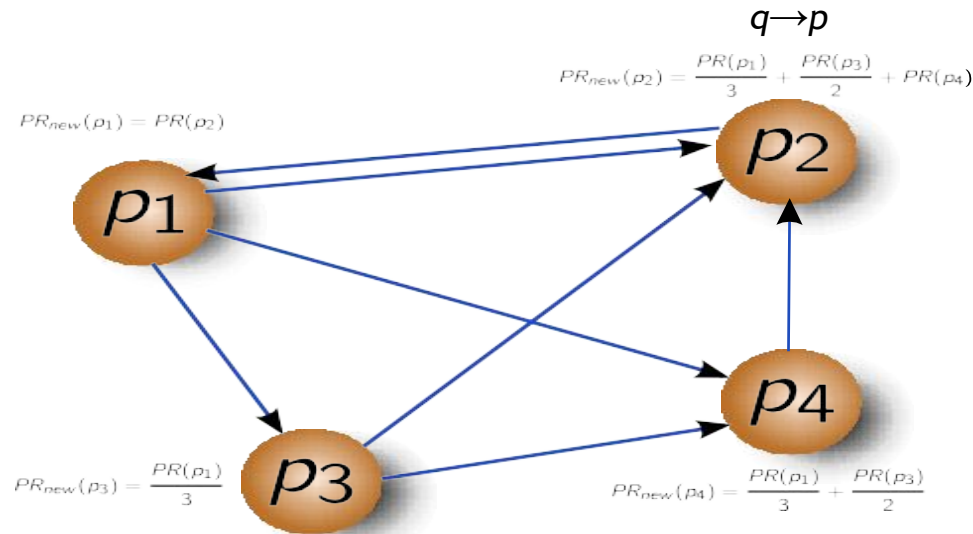
“Random surfer” clicks on successive links at random $\rightsquigarrow$ equal

- probability of each link to be clicked

Surfer gets bored and periodically jumps to another random page

PageRank Example

Example (set $\beta = 1$): $PR(p) = \frac{PR(q)}{out(q)}$



PageRank Calculation I

Iterative Computation of PageRank $PR(p)$ for a page p

$$PR(p) = \overbrace{(1 - \beta) \times \frac{1}{N}}^{\text{Surfer gets bored}} + \underbrace{\beta \times \sum_{q \rightarrow p} \frac{PR(q)}{out(q)}}_{\text{Randomly following links}}$$

N : Number of nodes / pages

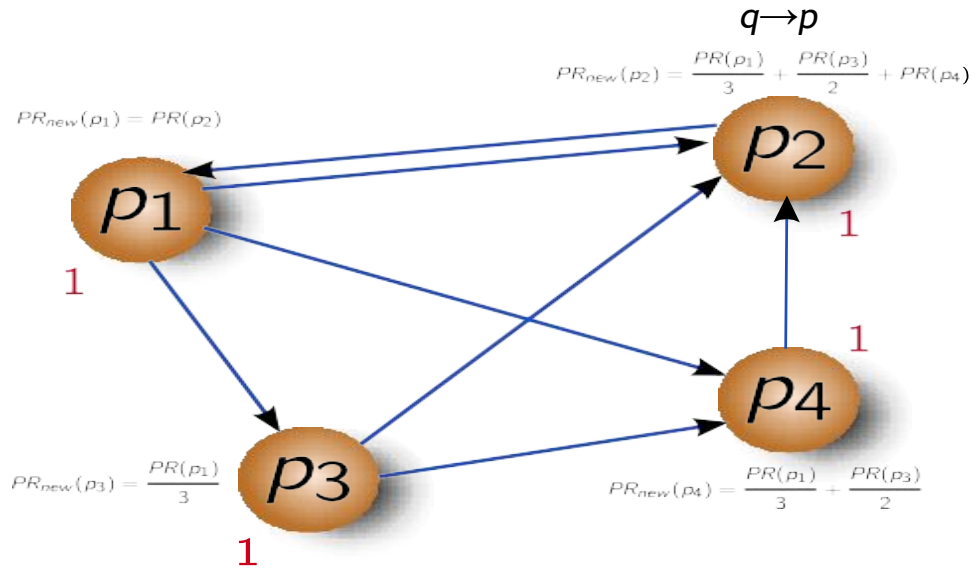
β : Constant between 0 and 1

$out(q)$: Number of outgoing links from page q

$q \rightarrow p$: There is a link from q to p

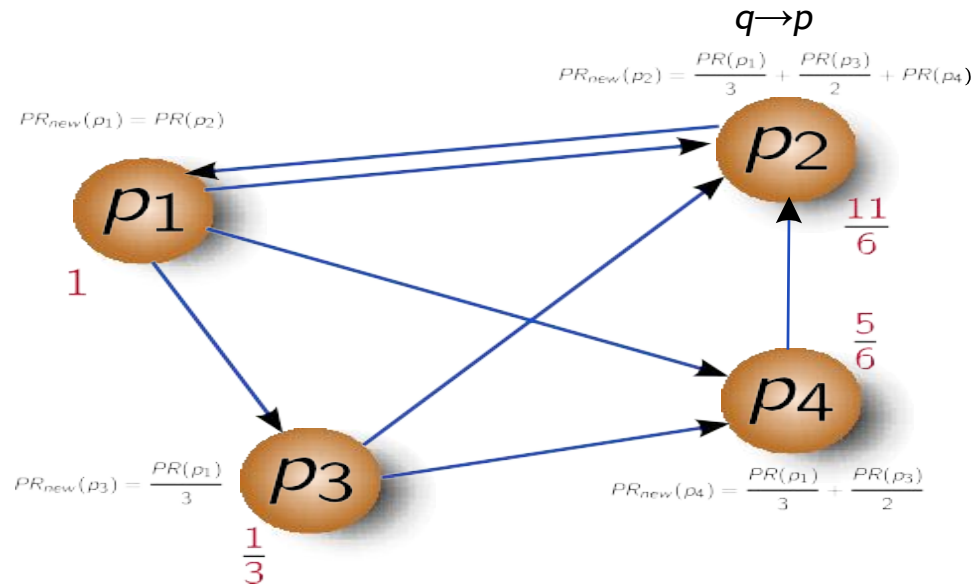
PageRank Example

Example (set $\beta = 1$): $PR(p) = \frac{PR(q)}{out(q)}$



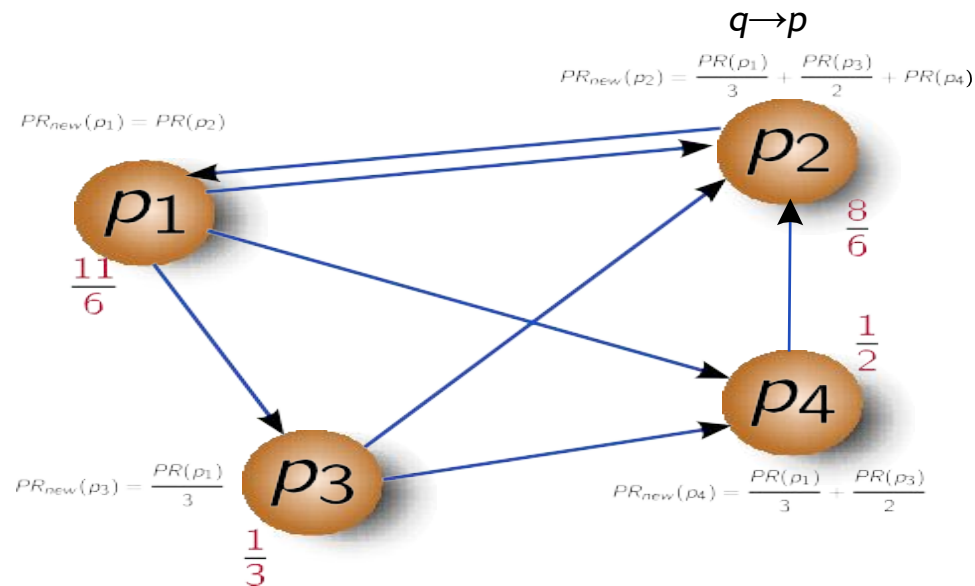
PageRank Example

Example (set $\beta = 1$): $PR(p) = \frac{PR(q)}{out(q)}$



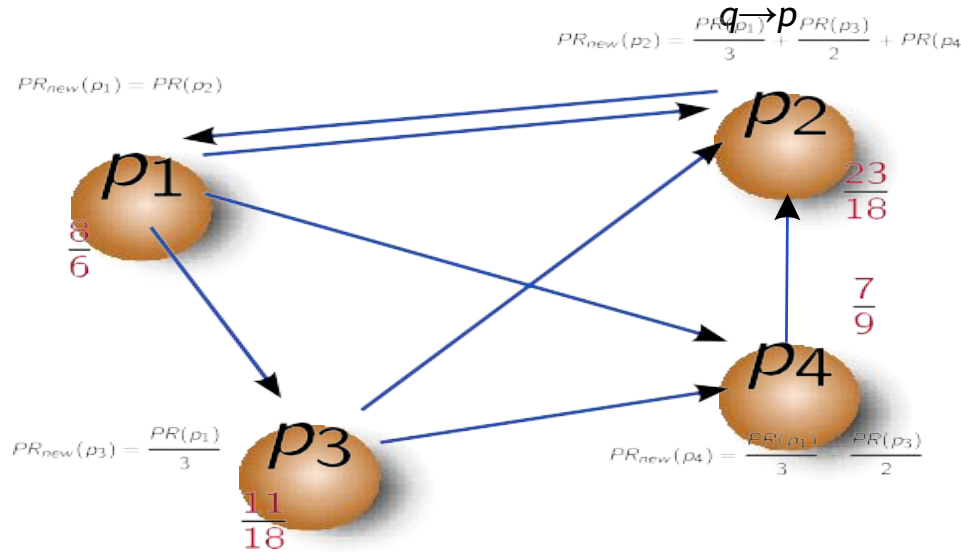
PageRank Example

Example (set $\beta = 1$): $PR(p) = \frac{PR(q)}{out(q)}$



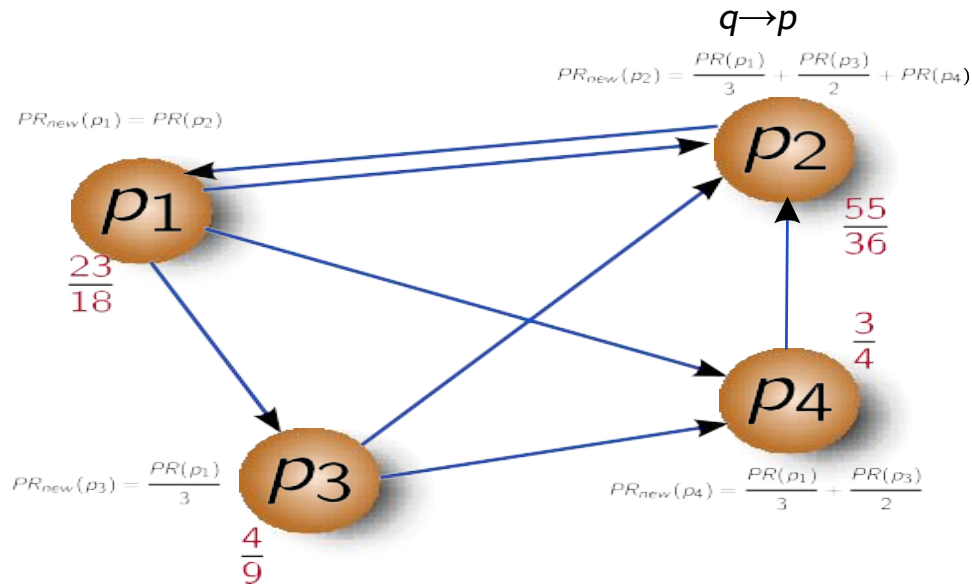
PageRank Example

Example (set $\beta = 1$): $PR(p) = \frac{PR(q)}{out(q)}$



PageRank Example

Example (set $\beta = 1$): $PR(p) = \frac{PR(q)}{out(q)}$



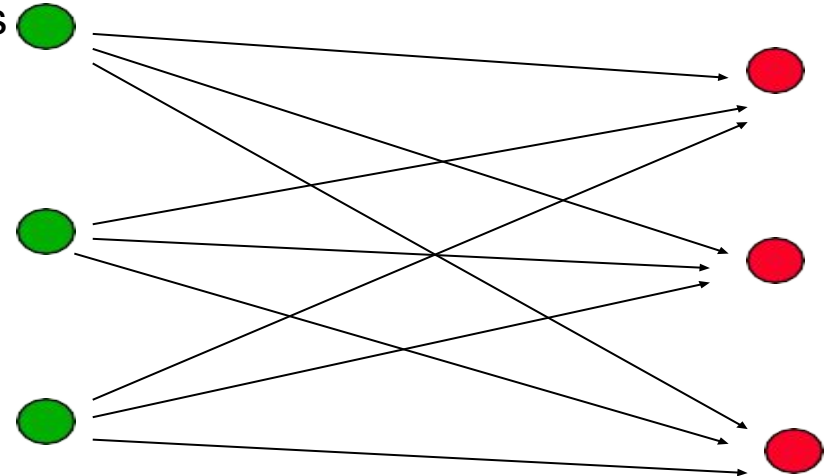
Kleinberg's HITS Algorithm

[[Kleinberg, 1998](#)]

Determining hubs and authorities of a certain topic

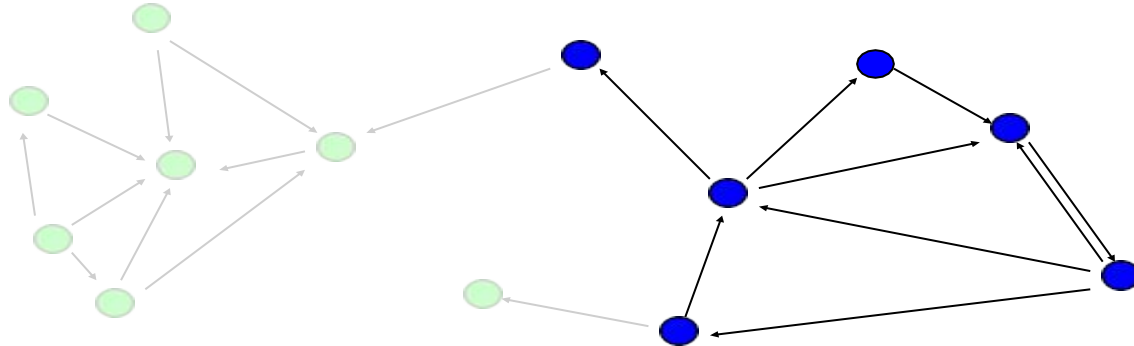
Hub Links to good authorities

Authority Has many links from good hubs



HITS: 2-step Algorithm

- 1 Compute the RSV for each Web site w.r.t. the query. Based on this, choose the subnet of relevant documents (**neighbourhood graph**)



- 2 Compute hub and authority values for each Web site in the neighbourhood graph

Hub And Authority Computation

Iterative Computation of authority a_p and hub value h_p

$$a_p = \sum_{q \rightarrow p} h_q$$

$$h_q = \sum_{q \rightarrow p} a_p$$

a_p : Authority weight for node/page p

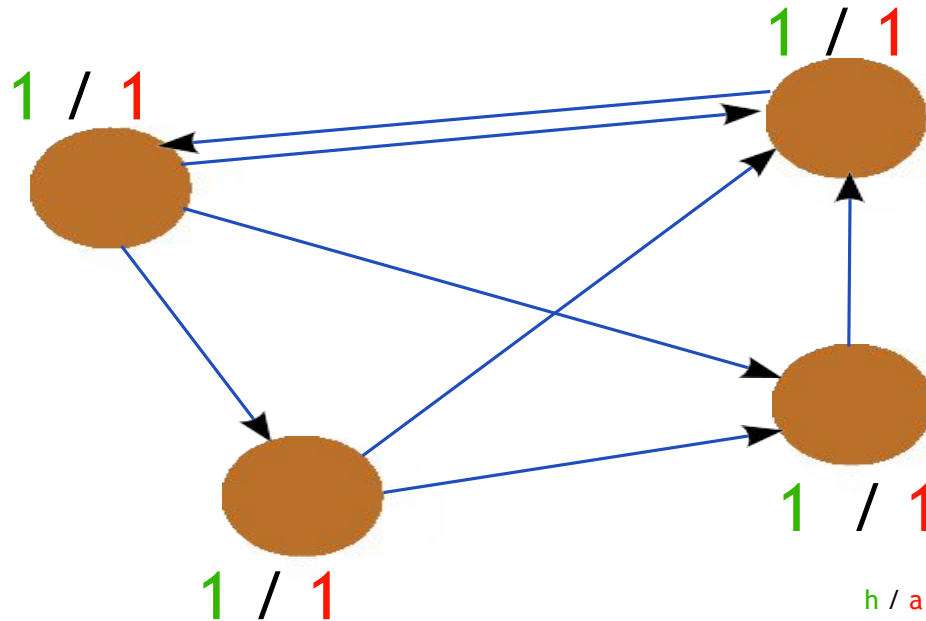
h_q : Hub weight for node/page q

and **normalisation criterium**

$$\sum_p (a_p)^2 = 1 \quad \text{and} \quad \sum_q (h_q)^2 = 1$$

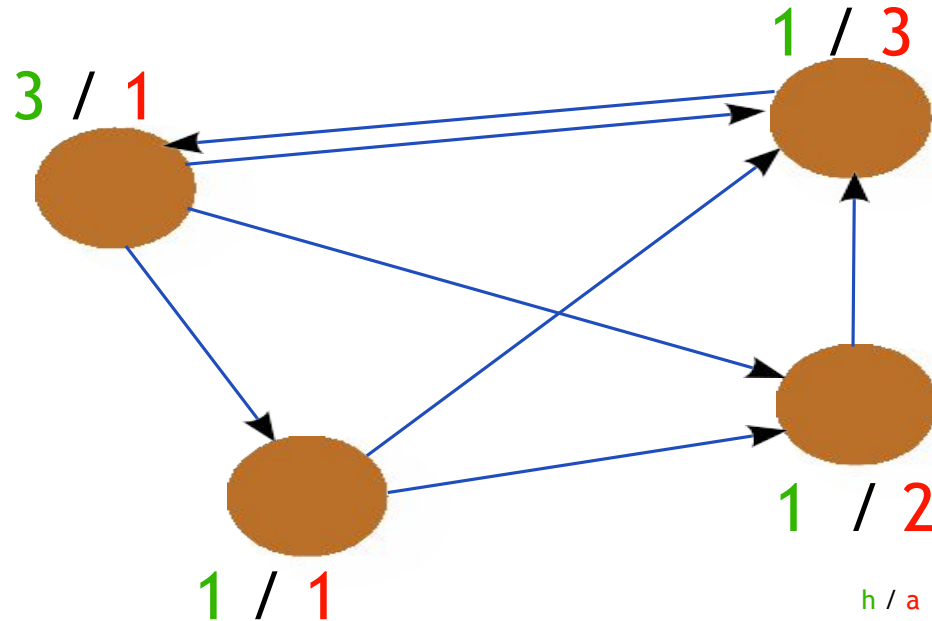
Start

HITS Example



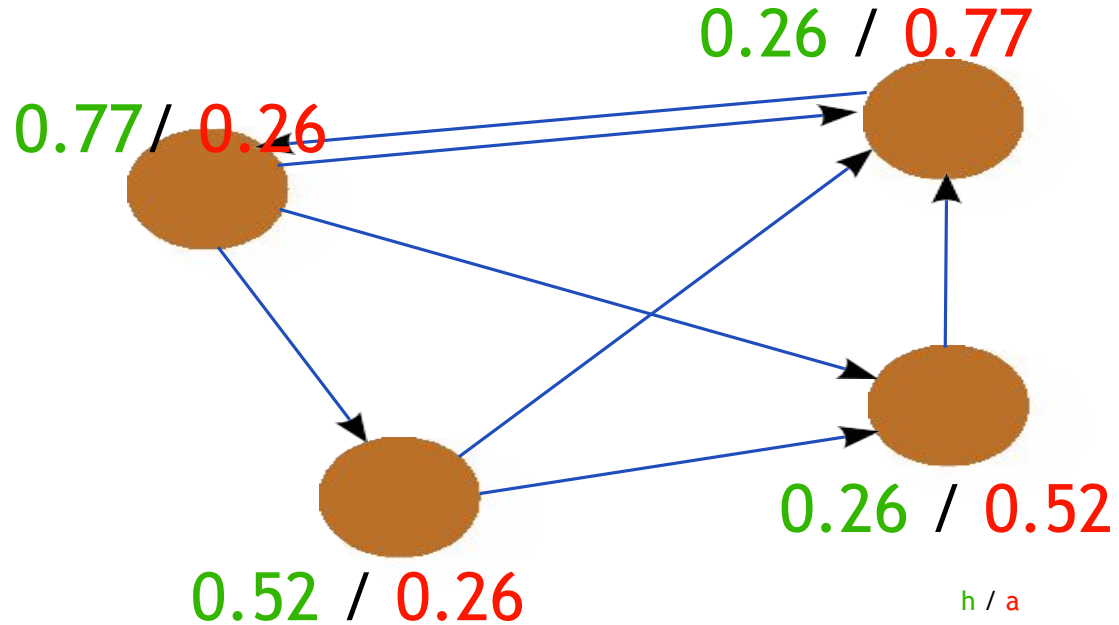
HITS Example

1st Iteration



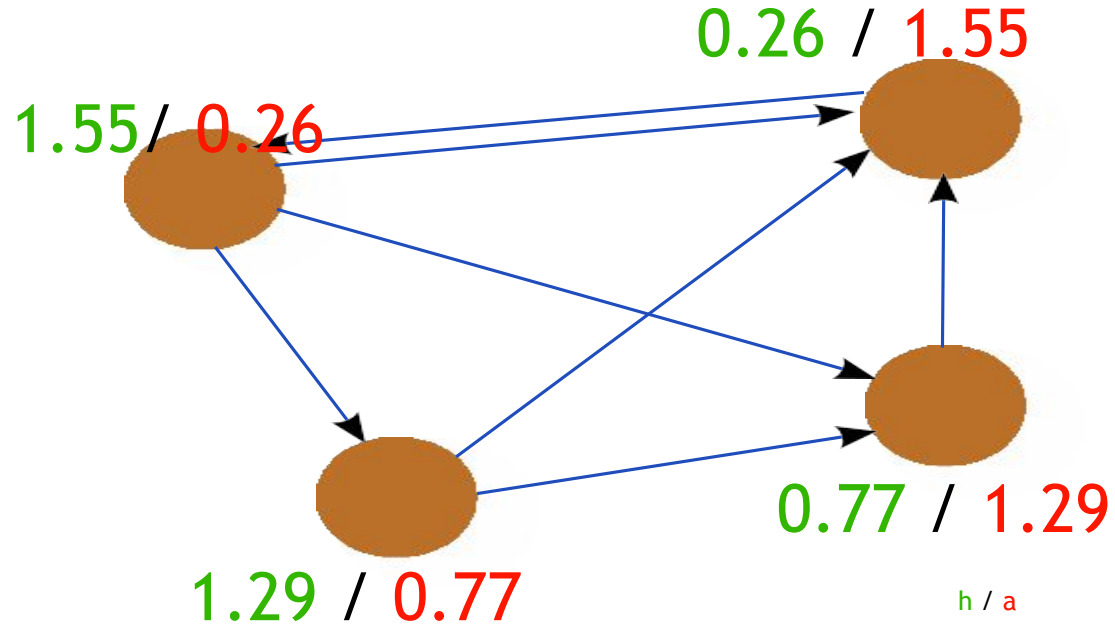
HITS Example

Normalisation



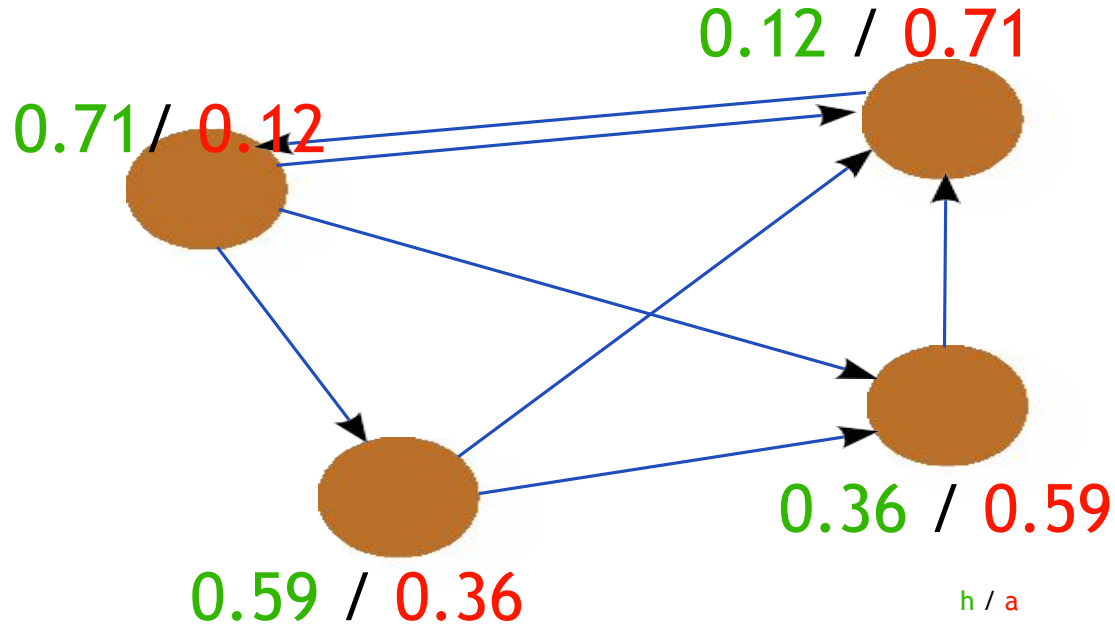
HITS Example

2nd Iteration



HITS Example

Normalisation



THE END